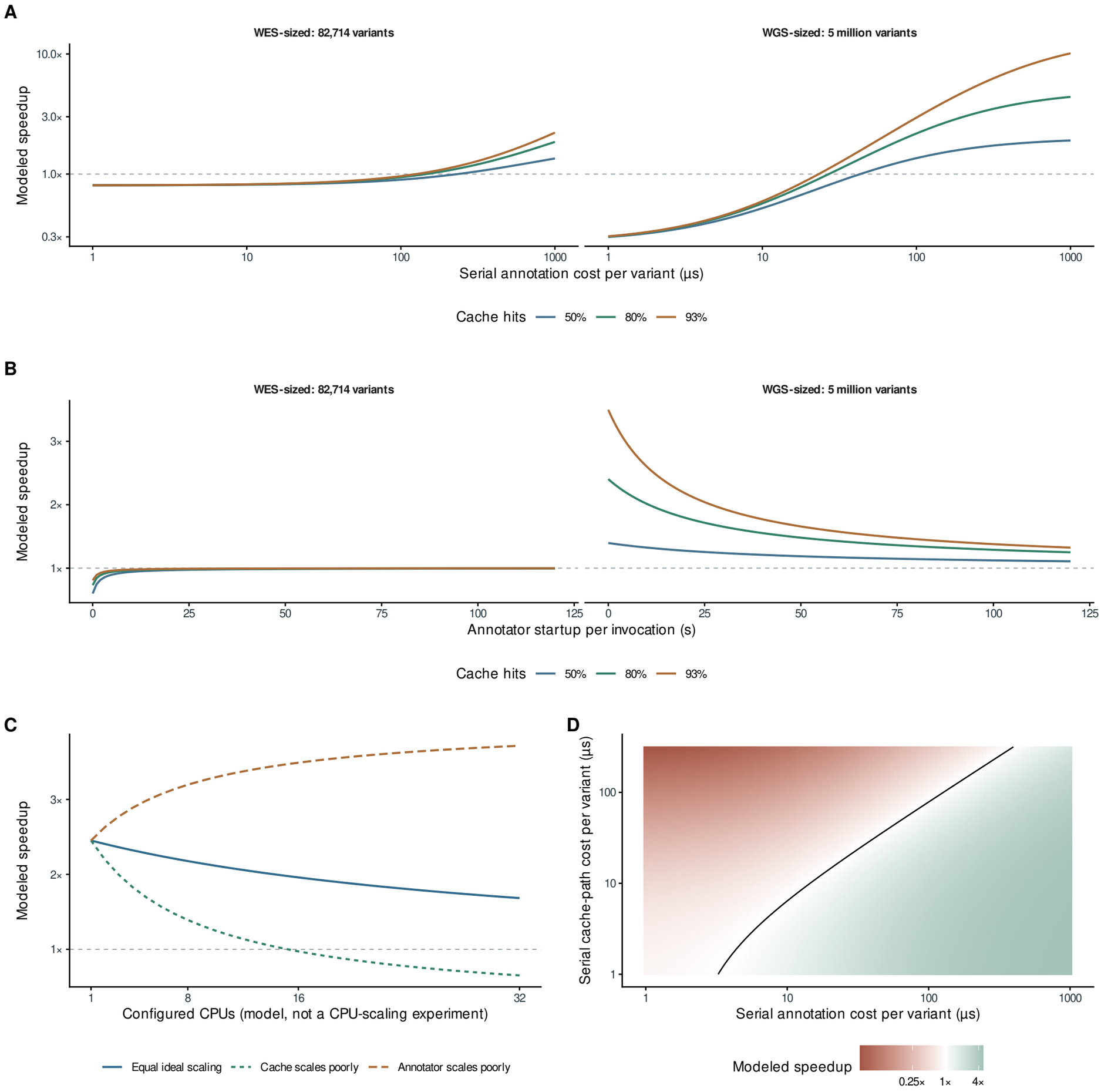
A
WES-sized: 82,714 variants
WGS-sized: 5 million variants
Modeled speedup
10.0×
3.0×
1.0×
0.3×
1
10
100
1000
Serial annotation cost per variant (µs)
Cache hits
50%
80%
93%
B
WES-sized: 82,714 variants
WGS-sized: 5 million variants
Modeled speedup
3×
2×
1×
0
25
50
75
100
125
Annotator startup per invocation (s)
Cache hits
50%
80%
93%
C
Modeled speedup
3×
2×
1×
1
8
16
32
Configured CPUs (model, not a CPU-scaling experiment)
Equal ideal scaling
Cache scales poorly
Annotator scales poorly
D
Serial cache-path cost per variant (µs)
100
10
1
1
10
100
1000
Serial annotation cost per variant (µs)
Modeled speedup
0.25×
1×
4×